

Deep Learning for teens

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Transfer learning

- ▶ We freeze all the layers except the last one.
- ▶ When we do that, the model's weights don't change during training.
- ▶ `base_model.trainable = False`

Transfer Learning vs Fine-Tuning

- ▶ Transfer Learning (Feature Extraction)
 - ▶ Freeze the pre-trained layers.
 - ▶ Train only a new classifier head.
 - ▶ Best for small datasets or when your task is similar to the original training data.
- ▶ Fine-Tuning
 - ▶ Start from pre-trained weights.
 - ▶ Unfreeze some or all layers and retrain them with your dataset.
 - ▶ Best for larger datasets or when your task is quite different from the original training data.

Transfer Learning vs Fine-Tuning

- ▶ Transfer learning: reuse a pre-trained model for a new task.
- ▶ Fine-tuning: a type of transfer learning where you unfreeze layers and retrain them.
- ▶ Key difference: transfer learning usually freezes most layers (feature extraction), while fine-tuning updates more weights for deeper adaptation.
- ▶ When to use:
 - ▶ Simple transfer learning → small, similar datasets.
 - ▶ Fine-tuning → larger, somewhat different datasets.
 - ▶ Train from scratch → vastly different datasets.

ImageNet

- ▶ It contains ~14 million labeled images across 1000 categories.
 - ▶ Animals
 - ▶ Vehicles
 - ▶ Food items
 - ▶ Musical instruments
 - ▶ Clothing & accessories
 - ▶ Household objects
 - ▶ Tools & devices
 - ▶ Outdoor scenes & natural objects
 - ▶ Indoor objects & furniture
 - ▶ Sports equipment & toys

ResNet50 model

- ▶ It is:
 - ▶ a model architecture (50 layers deep)
 - ▶ trained on the ImageNet dataset
 - ▶ used as a feature extractor in transfer learning

include_top=False

- ▶ Remove the original classifier from the model.
 - ▶ Don't give me the original final layer of the model.

GlobalAveragePooling2D vs AveragePooling2D

- ▶ AveragePooling2D (regular pooling)
 - ▶ Shrinks the feature map a little
 - ▶ Uses small windows (like 2×2 or 3×3)
 - ▶ Output is still 2-D
 - ▶ Example:
 - ▶ $28 \times 28 \rightarrow 14 \times 14 \rightarrow 7 \times 7$
- ▶ GlobalAveragePooling2D
 - ▶ Shrinks the feature map completely
 - ▶ Takes the average of the entire feature map
 - ▶ Output becomes 1 number per channel (1-D vector)
 - ▶ Example:
 - ▶ $7 \times 7 \times 2048 \rightarrow 2048$

Homework

- ▶ I want you guys to use the MobileNetV2 model with the same dog-and-cat dataset and repeat the steps I have done.